



9051 - EXploring the WIM with JWST MIRI

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The Warm Ionized Medium (WIM) is a major phase of the Galactic diffuse interstellar medium. Mid-IR fine structure emission lines, [NeII] 12.81 micron, [SIII] 18.71 micron and possibly [SIV] 10.51 micron, from the WIM were detected in the MIRI-MRS commissioning data. Compared to the traditional optical nebular lines the mid-IR fine structure lines are insensitive to extinction and the temperature of the WIM. Here we propose to analyze archival datasets ("background", faint targets, whether Galactic or extra-galactic) and using ionization ratios infer the spectrum of the diffuse Lyman-continuum radiation field as well as probe the field as a function of vertical height -- topics of considerable interest to the study of the halo and circumgalactic medium of the Milky Way.

OBSERVING DESCRIPTION